

Hard Gondaliya

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Master of Science in Computer Science

Sep. 2025 – May 2027

- Coursework: Natural Language Processing, Foundations of AI, Algorithms, Database Management Systems

NMIMS University

Shirpur, India

Bachelor of Technology in Computer Engineering

Jun. 2021 – May 2025

- Coursework: Machine Learning, Data Structures, Image & Video Processing, Computer Networks

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, R

LLMs & GenAI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Fine-Tuning, Transfer Learning, Hugging Face, LangChain, FAISS, Prompt Engineering

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, Convolutional Neural Networks (CNN), Transformers, Natural Language Processing (NLP), Hyperparameter Tuning, Model Quantization & Edge Deployment (Vitis AI, FPGA)

Cloud & MLOps: AWS (EC2, S3), Docker, MLflow, FastAPI, Git, Linux

Data & Analytics: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Feature Engineering, ETL Pipelines, Power BI, Tableau, Excel (Power Query, Pivot Tables)

Databases: PostgreSQL, MySQL, MongoDB

PROFESSIONAL EXPERIENCE

Space Applications Centre (SAC), ISRO

Ahmedabad, India

Machine Learning Intern

Jan. 2025 – May 2025

- Engineered a real-time CNN inference pipeline on FPGA for high-throughput satellite imagery at 30+ FPS, cutting latency by 40%
- Architected an integrated workflow spanning image reconstruction, preprocessing, and inference across 10,000+ images daily
- Deployed quantized models on FPGA hardware using Vitis AI, sustaining 98% accuracy under on-board resource limits

Sardar Vallabhbhai National Institute of Technology (SVNIT)

Surat, India

Machine Learning Engineer Intern

May 2024 – Jul. 2024

- Investigated plant disease detection on the PlantDoc dataset, evaluating 2,500+ annotated images for publication
- Trained deep neural networks to 95.8% peak validation accuracy using CLAHE and Local Binary Patterns (LBP) preprocessing

Prowiz Infotech LLP

Remote, India

Data Science Intern

Mar. 2024 – Apr. 2024

- Curated 50,000+ records with systematic preprocessing and quality checks, boosting downstream model precision by 15%
- Benchmarked 5+ machine learning algorithms while managing production data pipelines, cutting pipeline runtime by 20%

Inficial Infotech Pvt. Ltd.

Surat, India

Software Engineer Intern

May 2023 – Jul. 2023

- Developed a full-stack internal observability dashboard (React, FastAPI, PostgreSQL) tracking uptime and latency across 8 core microservices, with automated Slack alerts cutting incident discovery time by an estimated 15 minutes

PROJECTS

Terms of Service Risk Analyzer | Python, LLMs, Hugging Face, FAISS, FastAPI

2026

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system to parse dense user agreements, embedding legal text chunks with Hugging Face sentence-transformers into a FAISS vector index
- Integrated an LLM API with constrained prompt engineering to extract hidden privacy clauses and arbitration rules, generating grounded risk summaries with exact document citations
- Served the application via a FastAPI backend, containerized with Docker and deployed on AWS EC2, processing multi-page contracts in under 15 seconds

Transformer Language Model from Scratch | Python, PyTorch, NLP

2026

- Implemented a Transformer encoder and autoregressive language model from scratch in PyTorch, including multi-head self-attention, positional encodings, and causal masking
- Achieved 99.7% test accuracy on sequence classification and 5.30 perplexity on language modeling, tracking training runs and hyperparameters with MLflow

Waste Detection and Classification System | Python, YOLOv11, OpenCV

2024

- Constructed a real-time computer vision system with YOLOv11 achieving 94%+ detection accuracy, integrating conveyor-belt automated sorting and data augmentation for robustness

Toxic Comment Classification with Fine-Tuned DistilBERT | Python, Hugging Face, PyTorch

2026

- Fine-tuned DistilBERT on the Jigsaw toxic comment dataset using Hugging Face Transformers, benchmarking transfer learning performance against a from-scratch feedforward neural network baseline

RESEARCH & PUBLICATIONS

Comprehensive Review on Waste Management System Using ML and DL Algorithms

Springer, 2026

Gondaliya, H., et al. — *Lecture Notes in Networks and Systems, Vol. 1579 (ASSET 2025)*, peer-reviewed